

Overview and motivation :

Diabetes is a major and growing public health challenge in the United States and worldwide. Its strong connections to factors such as obesity, hypertension, smoking, age, and heart disease make it a complex condition that cannot be fully understood through raw data alone.

Organizations like the CDC and WHO have reported steadily increasing prevalence rates, emphasizing the need for tools that help reveal patterns, disparities, and risk factors in a clear and interpretable way.

Our project is motivated by the goal of using data visualization to explore these relationships. By analyzing demographic, lifestyle, and clinical variables together, we aim to highlight which populations are at higher risk, how indicators interact, and where notable trends emerge. Visualizations allow us to uncover insights that are not immediately visible in tabular form, making them well-suited for a dataset of this scope.

We use the *Comprehensive Diabetes Clinical Dataset* from Kaggle, which provides 100,000 records and 16 attributes ranging from demographic fields (age, gender, race) to lifestyle indicators (smoking history) and clinical metrics (BMI, HbA1c levels, blood glucose, hypertension, heart disease). The variety of attribute types such as categorical, binary, temporal, and continuous, makes the dataset ideal for building a dashboard with multiple coordinated views.

At least one visualization will present raw, unaggregated data (such as plotting HbA1c vs. blood glucose), while others will support comparisons, filtering, or geographic patterns. Together, these visualizations will help us communicate meaningful insights about diabetes risk factors while demonstrating thoughtful and effective visualization design.

Our dashboard leverages four primary views:

- A Parallel Coordinates Plot (PCP) for multivariate patient-level patterns
- A U.S. choropleth map showing state-level diabetes prevalence
- A Radial Bar Chart summarizing racial disparities
- A Radar Chart comparing smoking history distribution between diabetic and non-diabetic groups

Together, these visualizations reveal trends that are not obvious individually and enable deep exploratory analysis through coordinated interaction.

Related work:

Our visualization design draws inspiration from several sources, including academic research, real-world dashboards, and examples discussed in class. Much of our approach aligns with established principles in information visualization, particularly those emphasized by Edward Tufte and Tamara Munzner. Their recommendations showed data clearly, reducing unnecessary chartjunk, and leveraging appropriate visual channels for different attribute types, guided many of our layout and encoding decisions.

Scatterplots for medical variables, especially HbA1c and blood glucose, are commonly used in clinical research to illustrate metabolic trends. Existing diabetes studies often rely on similar plots to highlight correlations between short-term and long-term glucose indicators, reinforcing our choice of a raw scatterplot as a foundational visualization.

Heatmaps have also been widely used in public-health analytics to compare risk factors across demographic groups. Visualizations from the CDC's diabetes surveillance reports inspired our use of a grid-based representation for age and BMI, as these charts effectively reveal risk gradients across population segments.

Finally, the structure of our faceted bar charts was influenced by data-journalism outlets such as *The New York Times* and *Our World in Data*, which frequently use small-multiples layouts to compare trends across demographic slices. These examples demonstrated the value of separating gender and race into facet rows and columns to reduce clutter and maintain interpretability. Together, these sources helped shape our dashboard toward clarity, comparability, and meaningful storytelling with clinical data

Data source:

Kaggle- Diabetes Dataset

Exploratory data analysis:

Before designing the final dashboard, we performed a series of exploratory visualizations to better understand the structure of the dataset, identify potential patterns, and determine which variables would be most meaningful to visualize. These early explorations helped reveal relationships between clinical measurements, demographic attributes, and diabetes status, directly informing the final design choices.

1. Multivariate Clinical Trends

Initial scatterplots and correlation matrices showed:

- Strong separation between diabetic and non-diabetic groups in HbA1c and Blood Glucose
- BMI moderately correlated with diabetes but more overlapped
- Age had the weakest direct separation

This motivated using a Parallel Coordinates Plot rather than isolated scatterplots.

2. Geographic Patterns

Aggregating diabetes prevalence by state showed noticeable regional variation. This motivated the Choropleth Map, which reduces cardinality and offers spatial insight.

3. Race and Smoking History

Bar charts revealed:

- Clear differences in prevalence between racial groups
- Subtle but detectable differences in smoking distributions between diabetics and non-diabetics

These patterns motivated using:

- Radial Bar Chart for race
- Radar Chart for smoking history

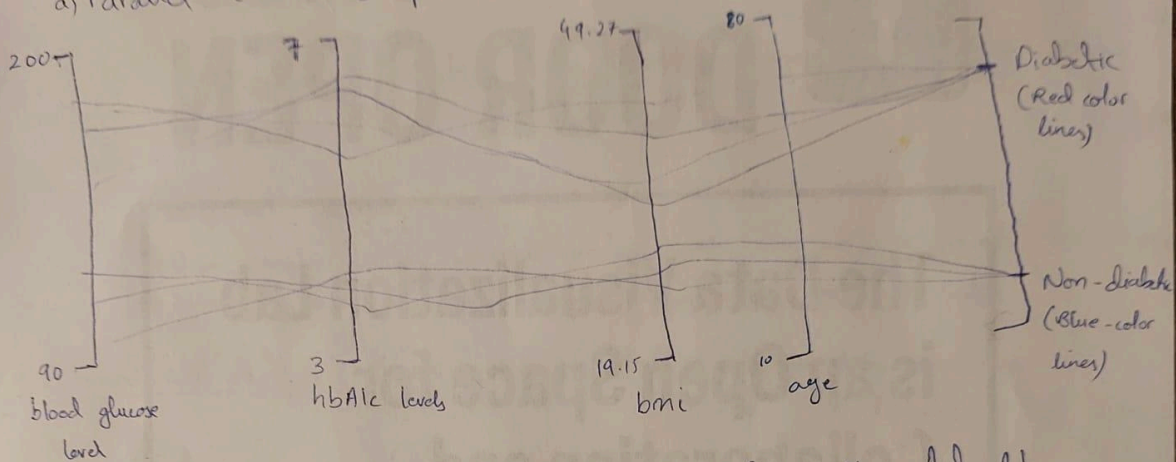
EDA directly shaped which views were included, what variables were highlighted, and how interaction should work.

Design Evolution:

Here is an overview of our design process and decisions we made along the way

1) Overview Visualizations

a) Parallel coordinates plot (PCP)



(Each line represents one row from the dataset)

(brush on each axis: when the user drags a brush range (say $HbA1c \geq 5$), we dim all lines outside that range).

2) Second chart - reducing the cardinality

a) Choropleth US map: Use diabetes prevalence in our sample per state:

$$\text{rate}_{(\text{state})} = \frac{\# \{ \text{diabetes} = 1 \text{ in state} \}}{\# \{ \text{rows in state} \}}$$

→ Clicking a state on the map filters/highlights only the lines from that state in the parallel coordinates plot.

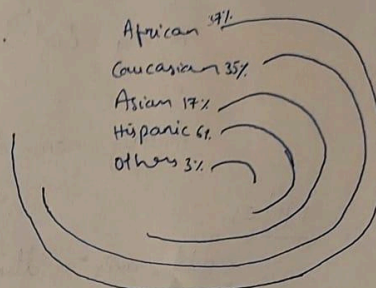
→ Conversely, we can also let brushing in the parallel coordinates plot recompute the per-state rates & recolor the map.

→ Tooltip on hover: (a) State name, (b) Diabetes prevalence %, (c) mean HbA1c for diabetics in that state, (d) mean BMI for diabetics in that state.

3) Smaller charts further reducing the cardinality

3.1) Radial Bar chart - Diabetes by Race

- Shows how diabetes prevalence varies across race groups.
- Each race category gets one arc segment.
- Clicking on arc segment filters the map & PCP for that particular race.



(Each bar will be colored differently for visual appeal)

Alternate design: A simple bar chart.

~~3.2) Stacked bar chart - Smoking History Composition~~

→ Shows

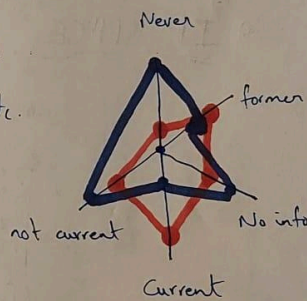
3.2) Radar Chart - Smoking History Composition

- five - six axes (based on categories) - Never, former, no info etc.

→ One red polygon = diabetics.

One blue polygon = non-diabetics.

- Interaction: ~~click segment/polygon~~ Click an axis → filters other charts to people with that smoking status.



Each axis = "composition by smoking history" for each outcome group.

So, → for the red (diabetic) polygon:

for each smoking category 'c':

$$\text{Value}_{\text{red}}(c) = \frac{\# \{ \text{diabetic patients with smoking} = c \}}{\# \{ \text{all diabetic patients} \}}$$

→ for the blue (diabetic) polygon:

~~for each~~

$$\text{Value}_{\text{blue}}(c) = \frac{\# \{ \text{non-diabetic patients with smoking} = c \}}{\# \{ \text{all non-diabetic patients} \}}$$

So each axis tells us:

Among diabetics, what fraction have this smoking status.

vs

Among non-diabetics, what fraction have this smoking status.

Notes:

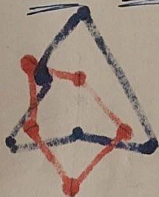
i) Add axis gridlines (e.g. 0%, 25%, 50%, 75%, 100%)

ii) Draw small circles at each vertex (refer diagram) so points are obvious

iii) Hovering over the circle should give the exact % value.

Alternate design: Simple bar chart
X-axis: smoking categories
Y-axis: % diabetic in that category.

3.3) IF SPACE EXISTS:



Two cards of Heart disease % by Gender & Hypertension % by Gender

Heart disease % by Gender	
Male:	27%
Female:	14%

Hypertension % by Gender	
Male:	32%
Female:	25%

(27% of men with heart disease among all men)

↳ Male bars would be blue & Female bars would be pink.

Clicking blue bars would filter other charts for only male stats

" pink " " " " " " " female "

~~Handwritten~~

Design Evolution

Our design evolved through multiple iterations:

Initial Concepts

We considered using:

- Heatmaps for age × BMI
- Faceted bar charts for categorical comparisons
- Scatterplots for clinical variables

However, these were removed due to:

- High overplotting
- Lack of multivariate context
- Redundancy with more effective visualizations

Final Visualization Set

We converged on four coordinated views:

1. Parallel Coordinates Plot (PCP) :patient-level multivariate trends
2. Choropleth Map:state-level prevalence
3. Radial Bar Chart: race-based prevalence comparison
4. Radar Chart: smoking distribution comparison

These views provide both micro-level and macro-level insight with smooth interactivity between them.

Implementation:

This section describes the intent, design, and interactivity of the four visualizations implemented in the project. Each visualization reflects a specific analytical goal: providing an overview of multivariate relationships, reducing cardinality by grouping, and enabling demographic subgroup comparison.

1. Parallel Coordinates Plot (PCP)

Intent:

The Parallel Coordinates Plot was designed as the primary overview visualization for the dataset. Because the dataset contains several important quantitative attributes relevant to diabetes diagnosis, Blood Glucose, HbA1c, BMI, and Age for which a multivariate visualization was needed to examine relationships and co-variation across these features. The PCP supports this by representing each patient as a polyline that spans the four aligned axes.

Functionality:

- Line Encoding:
Each line corresponds to one patient. Lines are colored based on diabetes status:
 1. Red = diabetic

- 2. Blue = non-diabetic
This enables immediate visual comparison of trends across outcome groups.
- Axes:
The axes represent:
 1. Blood Glucose Level
 2. HbA1c (%)
 3. BMI
 4. Age

These ranges match those shown in the sketches, with diabetic and non-diabetic lines crossing differently through these dimensions.
- Brushing Interaction:
Users can perform brushing on any axis. Dragging across a numeric range highlights only the lines whose values fall inside that range, dynamically dimming all other lines. This makes it possible to isolate specific subgroups, such as patients with HbA1c ≥ 7 or extremely high BMI.
- Coordinated Highlighting:
Brushing the PCP automatically filters and updates the other visualizations (map, radial bar chart, radar chart). This creates a continuous exploration environment where patterns detected in the PCP can be contextualized geographically or demographically.

2. U.S. Choropleth Map :State-Level Diabetes Prevalence

Intent:

The dataset contains many individual records, and patient-level visualization can become visually overwhelming. To reduce cardinality and introduce geographic context, we implemented a choropleth map that aggregates diabetes prevalence at the state level. This allows users to see statewide patterns and compare regional differences in the distribution of diabetes.

Functionality:

- Color Encoding:
Each state is shaded according to its diabetes prevalence]
Darker shades represent higher prevalence.
- Tooltip Details:
On hover, the map displays:
 1. State name
 2. Diabetes prevalence (%)
 3. Mean HbA1c among diabetics in the state
 4. Mean BMI among diabetics in the state
- These additional metrics provide richer context beyond binary prevalence.

- **State Selection Interaction:**
Clicking a state highlights only that state's patients in the PCP. All other lines are temporarily faded.
This enables users to examine how a state's population compares to the national multivariate patterns.
- **Reverse Interaction:**
Brushing a subset of patients in the PCP automatically recomputes and recolors state values, allowing users to see how the prevalence changes for brushed subsets.

Design Reference:

3. Radial Bar Chart : Diabetes Prevalence by Race

Intent:

The radial bar chart reduces categorical complexity and supports comparison between racial groups regarding diabetes prevalence. While a standard bar chart was considered as an alternative, the radial layout visually distinguishes categories in a more compact format and enhances group separation.

Functionality:

- **Structure:**
Each racial category is represented by a bar extending outward from a central point. The bar's length encodes the diabetes prevalence for that race.
- **Color Coding:**
Each race receives a distinct color to maintain clarity and improve quick scanning.
- **Filtering Interaction:**
Clicking on a race segment filters:
 - the PCP (only shows patients from that race),
 - the radar chart (recomputes smoking composition for that race),
 - and updates the map (if applicable).
- This enables focused analysis on specific demographic groups.

4. Radar Chart : Smoking History Composition

Intent:

The radar chart was designed to compare the distribution of smoking categories between diabetic and non-diabetic individuals. Unlike bar charts, radar charts emphasize the shape of each group's distribution, making it easier to visually detect similarities and differences across multiple categories.

Functionality:

- **Axes:**
Each axis corresponds to a smoking category present in the dataset (Never, Former,

Current, No Info).

- Polygons:
 - Red polygon: composition of diabetic patients.
 - Blue polygon: composition of non-diabetics.
The relative shape differences visually encode how the two groups differ or overlap.
- Filtering Interaction:
Clicking on any axis filters all other visualizations to only patients with that smoking status.
This allows the user to examine subpopulations (e.g., current smokers) in isolation.

Evaluation:

1. What We Learned from Using the Visualizations

PCP Findings

- Clear separation emerges between diabetics and non-diabetics on HbA1c and Blood Glucose, confirming these are the strongest diagnostic indicators.
- BMI differences are present but less visually distinct.
- Age is more mixed, suggesting it is not a strong standalone predictor in this dataset.

Choropleth Map Findings

- Diabetes prevalence varies meaningfully across states. Some regions show noticeably higher rates even after normalization.
- Using PCP brushing tied to the map reveals that high-HbA1c subgroups cluster in specific geographic regions.

Radial Bar Chart Findings

- Racial groups differ subtly in diabetes prevalence.
- The radial layout made patterns easier to compare at a glance, even with categories of uneven population sizes.

Radar Chart Findings

- Smoking history distributions differ between diabetic and non-diabetic groups, but not as dramatically as clinical variables.

- The diabetic polygon tends to have a slightly larger proportion of “Former” and “Current” smokers.
- Shape comparisons help highlight these small but potentially important differences.

2. How These Visualizations Helped Answer Analytical Questions

- Which variables most strongly correlate with diabetes?
PCP revealed that HbA1c and blood glucose dominate the separation.
- Do demographic factors such as race, geography, and smoking history play a role?
The coordinated views allowed demographic, geographic, and clinical variables to be explored together, revealing layered relationships.
- Can we isolate groups at higher risk?
Yes, interactions made it easy to isolate subpopulations, such as:
 - high-BMI individuals in high-prevalence states,
 - former smokers with elevated HbA1c,
 - specific racial groups with higher relative prevalence.

3. Visualization Effectiveness and Potential Improvements

Strengths

- Coordinated interaction greatly enhances exploration across multiple views.
- The PCP successfully provides multivariate context.
- The radial bar chart and radar chart efficiently summarize categorical variables.
- The map reduces cardinality and gives a geographic anchor for understanding prevalence.

Areas for Future Improvement

- Tooltips on the radial and radar charts would allow precise value inspection.
- PCP clutter could be reduced by adding opacity attenuation or curve bundling.
- Dynamic axis reordering in the PCP could reveal different patterns.
- Population-size normalization in the radial bar chart could make comparisons fairer.
- Map classification schemes (quantiles, natural breaks) could better highlight spatial variance.